

Dynamical incompatibilities in paced finger-tapping experiments

SUPPLEMENTAL MATERIAL

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A. Model simulations: incompatible vs compatible

In Figure S1 we show how model fitting to PS and SC from combined context smoothly accomodates all trajectories and the predicted asynchrony p_n doesn't show divergence. On the other hand, fittings to PS and SC from pure contexts show strong signs of incompatibility: transversal crossings in phase space, and divergent futures in predicted asynchrony. For simplicity, top row in Figure S1 (pure context) shows the same model simulations as Figure 2 in the main text.

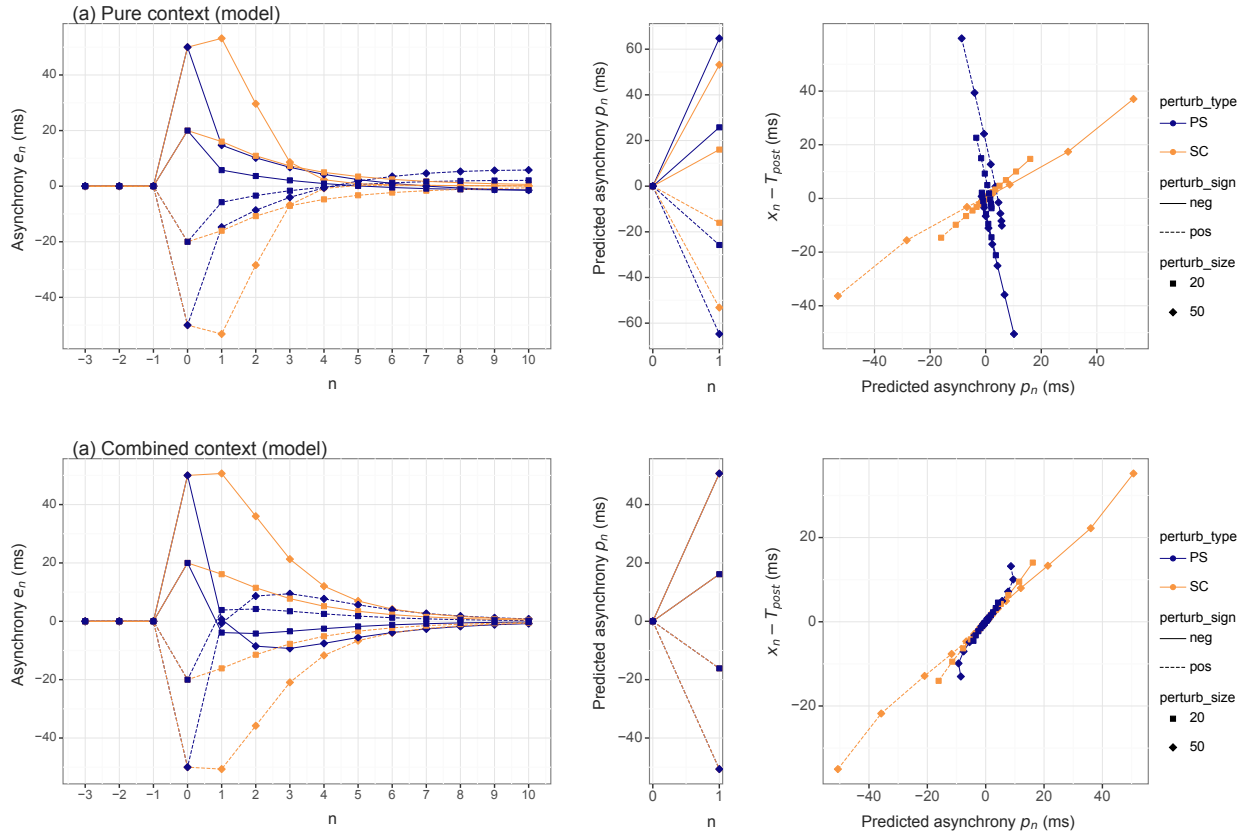


Figure S1. Model simulations. (a) Separate fitting to PS and SC from pure contexts. Left panel: asynchrony e_n time series. Middle panel: predicted asynchrony p_n showing divergent futures at $n = 1$ after sharing the same state at $n = 0$. Right panel: trajectories in model phase space show transversal crossings between PS (blue) and SC (orange) conditions. (b) Joint fitting to PS and SC from combined context. Predicted asynchrony p_n shows no divergence and trajectories in model phase space show no crossings but smooth merging.

B. Parameter joint distributions

Pairwise joint distributions of the fitted parameter values, obtained from joint fits of SC and PS using data from pure (Figure S2) and combined (Figure S3) contexts.

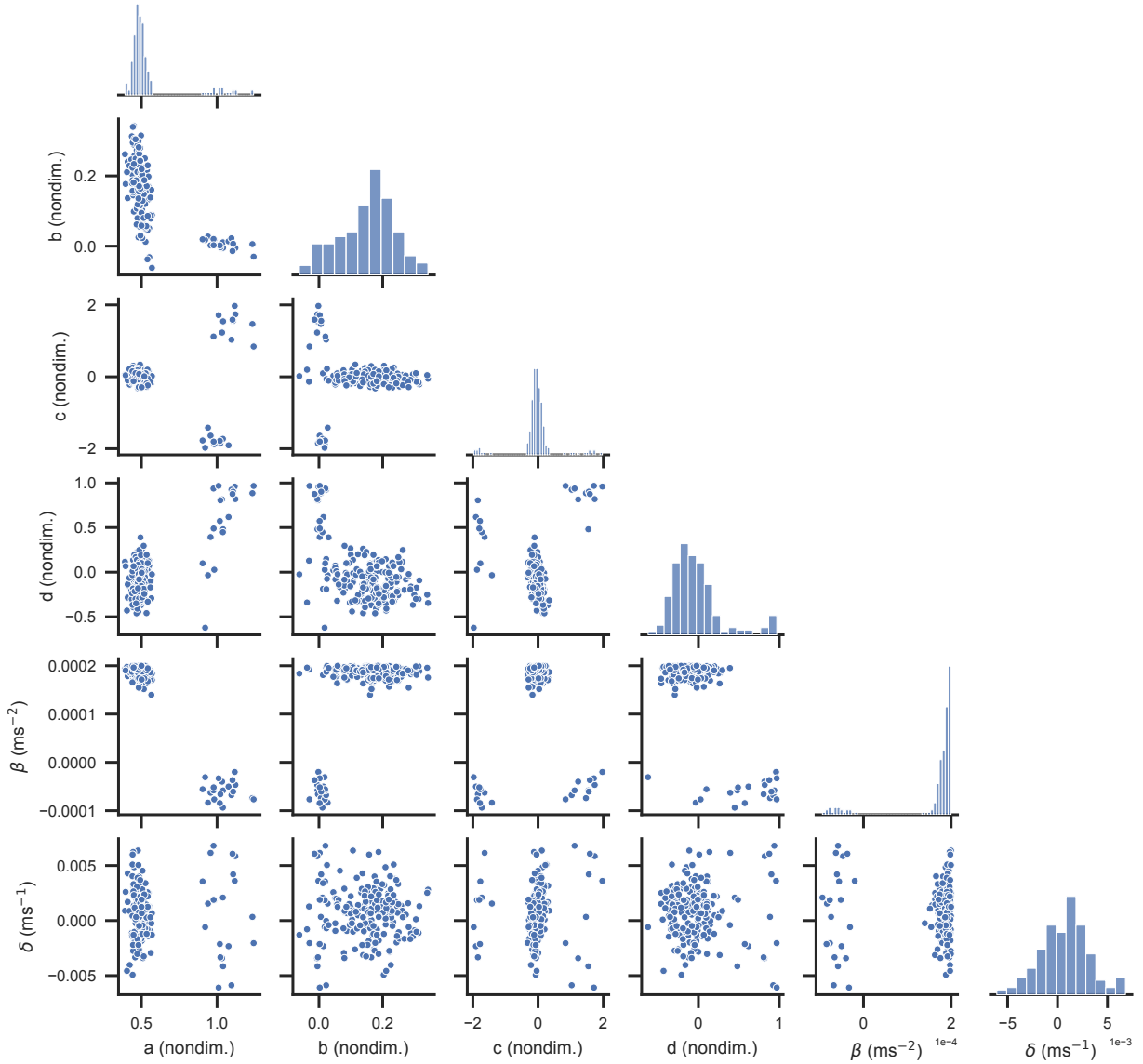


Figure S2. Pairwise joint distributions of fitted parameter values. Joint fitting of SC and PS from pure contexts.

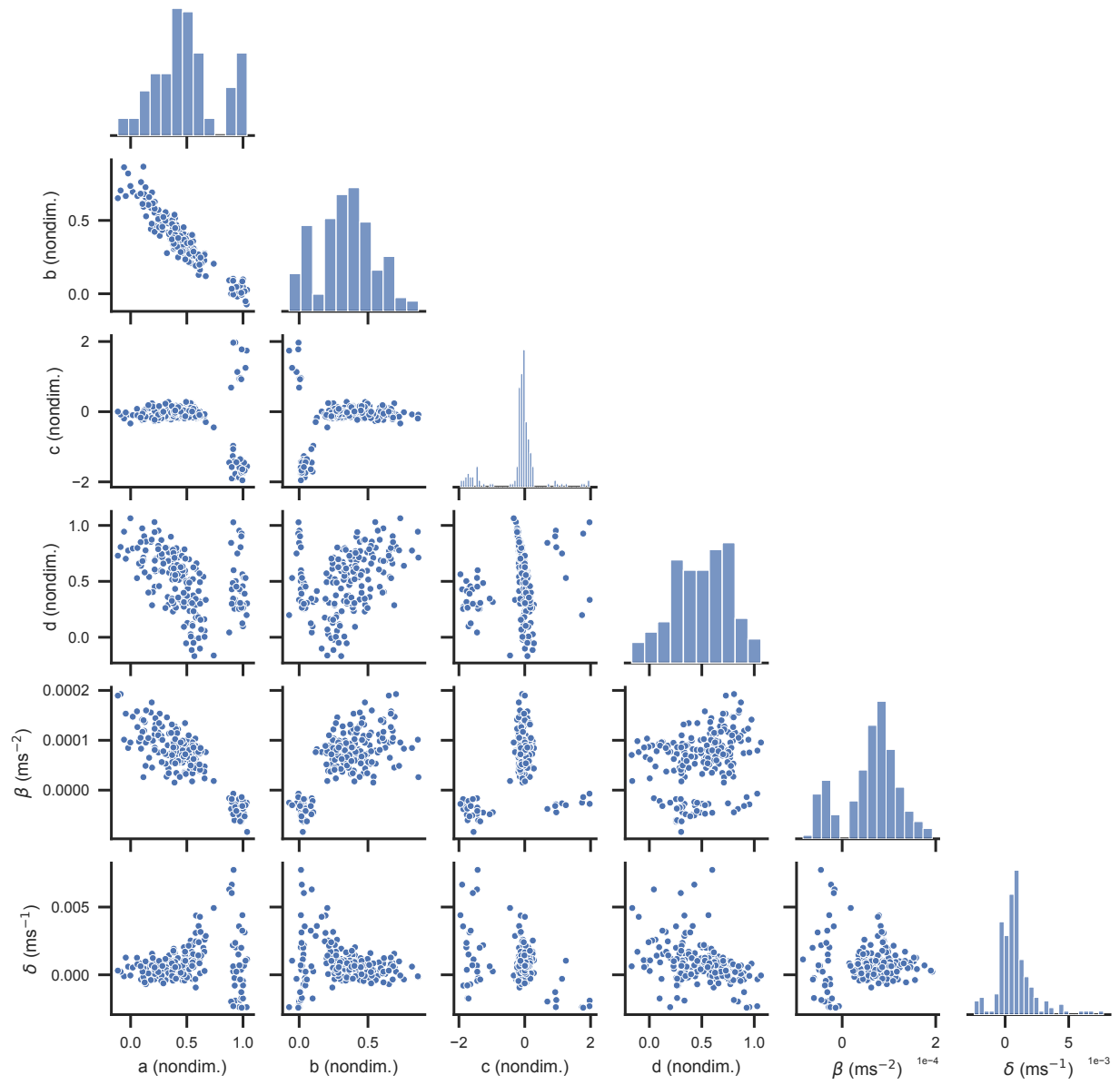


Figure S3. Pairwise joint distributions of fitted parameter values. Joint fitting of SC and PS from combined contexts.

C. Parameter correlations and incompatibilities

Here we show that parameter interdependence (correlation) is not an issue when concluding about incompatibility between models based on the non-overlapping obtained distributions.

Consider strictly correlated parameters A and B ; that is, A can be substituted by a

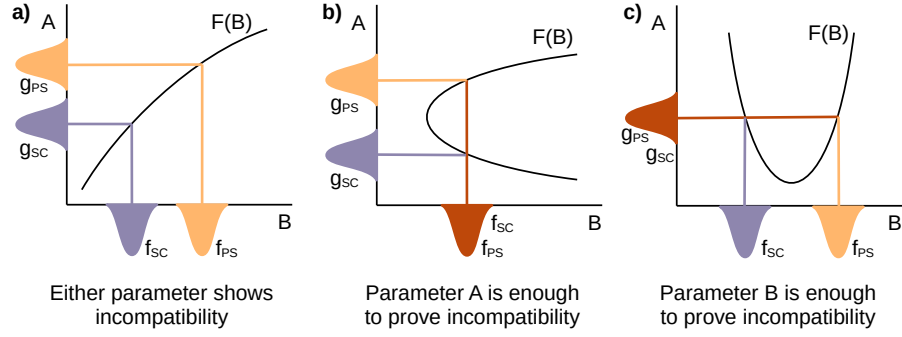


Figure S4. Correlation between parameters (parameter interdependence) is not an issue when concluding about incompatibility of solutions. (a) Monotone, single-valued correlation between parameters A and B . (b) Monotone, multi-valued correlation. (c) Non-monotone, single-valued correlation.

function of B : $A = F(B)$. Consider the comparison between SC and PS in pure contexts leading to incompatibility, and assume that the obtained distributions of parameter B are f_{SC} and f_{PS} , respectively. The corresponding distributions of parameter A are g_{SC} and g_{PS} . For the sake of this analysis, consider the following three scenarios depicted in Figure S4:

1. Function F is monotone and single-valued (Figure S4a). If f_{SC} and f_{PS} are non-overlapping, then g_{SC} and g_{PS} are also non-overlapping. Then the incompatibility arises from either parameter. That is, we will find incompatibility either in both parameters or none of them.
2. Function F is monotone and multi-valued (Figure S4b). If f_{SC} and f_{PS} are overlapping, it could still be the case that g_{SC} and g_{PS} are non-overlapping. In this case, parameter A would be enough to prove incompatibility.
3. Function F is non-monotone (Figure S4c). If f_{SC} and f_{PS} are non-overlapping, then g_{SC} and g_{PS} could be overlapping. In this case, parameter B would be enough to prove incompatibility.

Thus, regardless of the particular form of the potential correlation between parameters A and B , any sign of non-overlapping distributions of either parameter would suffice to prove incompatibility.